Homework 3 — changed numbers (extra practice)

Module 3 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

An antibiotic with a clearance of 4.62 L/hr and an apparent volume of distribution of approximately 30 L is to be administered by IV infusion to attain a steady-state plasma concentration of 15 mcg/mL.

a Module 3

What is an appropriate rate of infusion to achieve the desired steady-state concentration of 15 mcg/mL?

b Module 3

What is the half-life of the drug that is to be infused?

c Module 3

What is the expected concentration of drug in the plasma 6 hours after the start of the infusion?

d Module 3

How much time is required to reach 95% of the steady-state concentration if no loading dose were used?

e Module 3

Recommend a loading dose to obtain the steady-state concentration immediately.

f Module 3

If the infusion were stopped after the desired steady-state concentration of 15 mcg/mL was attained, what would the concentration be 8 hours after the cessation?

g Module 3

If the infusion rate that you recommended in a (above) were doubled, what would the new steady-state concentration be?

h Module 3

If the infusion rate that you recommended in a (above) were doubled, how long would it take to reach 95% of the new steady-state concentration (no loading dose)?

i Module 3

If the infusion were stopped after proceeding long enough at the rate determined in part g (above) to reach the new steady-state concentration, what would the concentration be 6 hours after the cessation?

j Module 3

What specific change would you make to achieve a steady state concentration of 9 mg/L? Give the new infusion rate.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

a

Worked by the drill

Answer: 69.3 mg/hr

WHAT THE STEM GIVES

Cl	4.62 L/hr	multiplied by C_{ss}
V_D	approximately 30 L	*not needed: the rate at steady state needs clearance only*
C_{ss}	15 mcg/mL	written 15 mg/L, multiplied by clearance

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = C_{ss} \times Cl = (15\ mg/L)(4.62\ L/hr) = 69.30\ mg/hr$$

At steady state the infusion rate R equals the rate of elimination, Cl × C_{ss}, where Cl is clearance and C_{ss} the steady-state concentration; $C_{ss} = \frac{R}{Cl}$ rearranges to $R = C_{ss} \times Cl$. 15 mcg/mL is 15 mg/L, and (mg/L)(L/hr) cancels the litres, leaving mg/hr: (15 mg/L)(4.62 L/hr) = 69.30 mg/hr.

> **DOES THE NUMBER MAKE SENSE?**
>
> Each hour 4.62 L of plasma is cleared; keeping 15 mg in each litre needs 15 × 4.62 = 69.30 mg/hr, 15 times the clearance. The rate is above zero.

Her homework wording with different numbers, so this is not the graded answer.

b

Worked by the drill

Answer: 4.5 hr

WHAT THE STEM GIVES

Cl	4.62 L/hr	the numerator of $k = \frac{Cl}{V_D} = 0.154\ hr^{-1}$
V_D	approximately 30 L	the denominator of k
C_{ss}	15 mcg/mL	*not needed: half-life is a property of the drug and patient*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{Cl}{V_D} = \frac{4.62\ L/hr}{30\ L} = 0.154\ hr^{-1}$$

Clearance equals k times the volume, so dividing clearance by volume gives the rate constant in reciprocal hours.

ALGEBRA

$$t½ = \frac{0.693}{0.154\ hr^{-1}} = 4.50\ hr$$

$t½ = \frac{0.693}{k}$, because $k\ t½ = \ln 2$ when the concentration falls to half; with $k = 0.154\ hr^{-1}$ the reciprocal hours invert to hours, 4.50 hr. Elimination is a property of the drug and the patient, so an infusion does not change k or the half-life; it only sets the level reached.

> **DOES THE NUMBER MAKE SENSE?**
>
> $k = \frac{4.62}{30} = 0.154\ hr^{-1}$ removes about 14% an hour, so halving takes between 4 and 5 hours: 4.50 hr.

Her homework wording with different numbers, so this is not the graded answer.

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c Worked by the drill

Answer: 9.05 mg/L

WHAT THE STEM GIVES

Cl	4.62 L/hr	gave $k = 0.154\ hr^{-1}$ in part b, the exponent
V_D	approximately 30 L	the denominator of k
C_{ss}	15 mcg/mL	the plateau R over Cl = 15 mg/L the fraction is applied to
t	6 hours after the start	t in $1 - e^{-kt}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = \frac{R}{Cl}(1 - e^{-kt}) = 15\ mg/L \times (1 - e^{-(0.154)(6)})$$

During a constant infusion the concentration is $C = \frac{R}{Cl}(1 - e^{-kt})$; R over Cl is the plateau C_{ss}, here 15 mg/L, and $(1 - e^{-kt})$ is the fraction of it reached by time t, because elimination has not yet grown to balance the input. $k = 0.154\ hr^{-1}$ from part b.

ALGEBRA

$$C = 15\ mg/L \times (1 - 0.3969) = 9.05\ mg/L$$

$kt = (0.154\ hr^{-1})(6\ hr) = 0.924$, a pure number, and $e^{-0.924} = 0.3969$ is the fraction of C_{ss} still missing; $1 - 0.3969 = 0.6031$ has been reached. Six hours is 1.33 half-lives of 4.50 hr, so the level is $15 \times 0.6031 = 9.05$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
>
> Six hours is between one and two half-lives of the drug: after one half-life 50% of the plateau is reached, after two 75%, so the level lies between half the plateau and three-quarters of it: 9.05 mg/L, below 15.

Her homework wording with different numbers, so this is not the graded answer.

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d

Worked by the drill

Answer: 19.45 hr

WHAT THE STEM GIVES

Cl	4.62 L/hr	gave k = 0.154 hr^{-1} in part b, the denominator
V_D	approximately 30 L	the denominator of k
C_{ss}	15 mcg/mL	*not needed: the fraction 0.95 is dimensionless*
fraction	95%	sets $e^{-kt} = 0.05$, so kt = ln(20) = 2.9957

THE WORKING, ONE LINE AT A TIME

SETUP

$$0.95 = 1 - e^{-kt} \rightarrow e^{-kt} = 0.05 \rightarrow t = \frac{\ln(20)}{k}$$

The fraction of C_{ss} reached during an infusion is $\frac{C}{C_{ss}} = 1 - e^{-kt}$, which depends only on kt. Setting it to 0.95 gives $e^{-kt} = 0.05$, so $kt = -\ln 0.05 = \ln 20 = 2.9957$ and $t = \frac{\ln(20)}{k}$; the infusion rate R cancels and plays no part.

ALGEBRA

$$t = \frac{2.9957}{0.154\ hr^{-1}} = 19.45\ hr\ (4.32\ half\text{-}lives)$$

ln 20 = 2.9957 is a pure number, so dividing by k = 0.154 hr^{-1} leaves hours: 19.45 hr. Because $k = \frac{0.693}{t½}$, the same time is $\frac{2.9957}{0.693}$ = 4.32 half-lives of the 4.50 hr from part b; without a loading dose, 95% of C_{ss} takes 4.32 half-lives at any infusion rate.

> **DOES THE NUMBER MAKE SENSE?**
> Each half-life closes half the remaining gap: 50, 75, 87.5, about 94% after four, so 95% needs a little more than four half-lives: 4.32 × t½ = 19.45 hr, between 4 and 5 half-lives.

Her homework wording with different numbers, so this is not the graded answer.

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e

Worked by the drill

Answer: 450 mg

WHAT THE STEM GIVES

Cl	4.62 L/hr	*not needed: a loading dose fills the volume; clearance sets the rate*
V_D	approximately 30 L	multiplied by C_{ss}
C_{ss}	15 mcg/mL	written 15 mg/L, multiplied by V_D

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = C_{ss} \times V_D = (15\ mg/L)(30\ L) = 450\ mg$$

The loading dose puts the steady-state amount in the body at once: concentration times volume gives an amount.

DOES THE NUMBER MAKE SENSE?
To have 15 mg in each of 30 litres at once takes 15 × 30 = 450 mg. The dose is above zero.

Her homework wording with different numbers, so this is not the graded answer.

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f

Worked by the drill

Answer: 4.38 mg/L

WHAT THE STEM GIVES

Cl	4.62 L/hr	gave k = 0.154 hr^{-1} in part b, the exponent
V_D	approximately 30 L	the denominator of k
C_{ss}	15 mcg/mL	the level the decay starts from
t	8 hours after the cessation	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = C_{ss}\, e^{-kt} = 15 \text{ mg/L} \times e^{-(0.154)(8)}$$

Once the infusion stops nothing more goes in, so the plasma level follows the one-compartment decline $C = C_0\, e^{-kt}$ with the concentration at the moment of stopping, C_{ss} = 15 mg/L, as the starting point; k = 0.154 hr^{-1} from part b and t = 8 hr counted from the cessation.

ALGEBRA

$$C = 15 \times 0.2917 = 4.38 \text{ mg/L}$$

kt = (0.154 hr^{-1})(8 hr) = 1.232, a pure number, and $e^{-1.232}$ = 0.2917 is the fraction of C_{ss} remaining: 15 × 0.2917 = 4.38 mg/L. Eight hours is 1.78 half-lives of 4.50 hr, so a little over a quarter of C_{ss} is left.

DOES THE NUMBER MAKE SENSE?
Eight hours is a little under two half-lives, so a little over a quarter of 15 mg/L remains: 4.38 mg/L, between 0.25 × 15 and 0.5 × 15.

Her homework wording with different numbers, so this is not the graded answer.

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g

Worked by the drill

Answer: 30 mg/L

WHAT THE STEM GIVES

Cl	4.62 L/hr	the denominator, unchanged
V_D	approximately 30 L	*not needed: C_{ss} depends on rate and clearance only*
C_{ss}	15 mcg/mL	the plateau at the original rate; doubles with the rate
R	doubled	2R = 138.60 mg/hr, the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{2R}{Cl} = \frac{138.60 \text{ mg/hr}}{4.62 \text{ L/hr}} = 30 \text{ mg/L}$$

$C_{ss} = \frac{R}{Cl}$: at steady state the input R equals the elimination Cl × C_{ss}, so the plateau is proportional to the rate at a fixed clearance. Doubling R from 69.30 to 138.60 mg/hr with Cl still 4.62 L/hr doubles C_{ss} from 15 to 30 mg/L; the litres per hour cancel to leave mg/L.

DOES THE NUMBER MAKE SENSE?

C_{ss} is proportional to the rate at a fixed clearance, so doubling R doubles the plateau from 15 to 30 mg/L; the time to reach it does not change.

Her homework wording with different numbers, so this is not the graded answer.

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h

Worked by the drill

Answer: 19.45 hr

WHAT THE STEM GIVES

Cl	4.62 L/hr	gave k = 0.154 hr^{-1} in part b, the denominator of ln(20) over k
V_D	approximately 30 L	the denominator of k
C_{ss}	15 mcg/mL	*not needed: R cancels out of the fraction reached*
R	doubled	*not needed: the time to a fraction of steady state depends on k alone*
fraction	95%	sets kt = ln(20)

THE WORKING, ONE LINE AT A TIME

SETUP

$$t = \frac{\ln(20)}{k} = 19.45 \text{ hr, the same as part d}$$

The time to a fraction of steady state depends only on k, so a faster rate reaches a higher plateau in the same time.

DOES THE NUMBER MAKE SENSE?

The fraction of the plateau reached is $1 - e^{-kt}$, with no R in it, so the doubled rate reaches 95% of its higher plateau in the same number of half-lives: 19.45 hr, as in part d.

Her homework wording with different numbers, so this is not the graded answer.

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i

Worked by the drill

Answer: 11.91 mg/L

WHAT THE STEM GIVES

Cl	4.62 L/hr	gave k = 0.154 hr^{-1} in part b, the exponent
V_D	approximately 30 L	the denominator of k

C_{ss}	15 mcg/mL	doubled in part g to 30 mg/L, the level the decay starts from
t	6 hours after the cessation	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = (30 \text{ mg/L})\, e^{-(0.154)(6)} = 30 \times 0.3969 = 11.91 \text{ mg/L}$$

After the infusion stops the level follows $C = C_0\, e^{-kt}$ from the concentration at cessation, the new plateau of 30 mg/L from part g. $k = 0.154$ hr^{-1} is unchanged, so $kt = 0.924$ and $e^{-0.924} = 0.3969$ of the plateau remains: $30 \times 0.3969 = 11.91$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is between one and two half-lives, so between a quarter and a half of the 30 mg/L plateau remains: 11.91 mg/L, between 0.25×30 and 0.5×30.

Her homework wording with different numbers, so this is not the graded answer.

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j

Worked by the drill

Answer: 41.58 mg/hr

WHAT THE STEM GIVES

Cl	4.62 L/hr	multiplied by the new target
V_D	approximately 30 L	*not needed: the rate at steady state needs clearance only*
C_{ss}	15 mcg/mL	the old target, replaced by 9 mg/L
new C_{ss}	9 mg/L	multiplied by clearance

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Change the infusion rate: } R = C_{ss} \times Cl = (9 \text{ mg/L})(4.62 \text{ L/hr}) = 41.58 \text{ mg/hr}$$

$C_{ss} = \dfrac{R}{Cl}$, and Cl, the clearance, is a property of the patient that cannot be prescribed, so the infusion rate R is the variable to change. For $C_{ss} = 9$ mg/L, $R = C_{ss} \times Cl = (9 \text{ mg/L})(4.62 \text{ L/hr}) = 41.58$ mg/hr, the litres cancelling; the rate falls from 69.30 mg/hr.

> **DOES THE NUMBER MAKE SENSE?**
> The target drops from 15 to 9 mg/L, three-fifths of before, so the rate drops to three-fifths of the part a rate: $9 \times 4.62 = 41.58$ mg/hr, below it.

Her homework wording with different numbers, so this is not the graded answer.
